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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

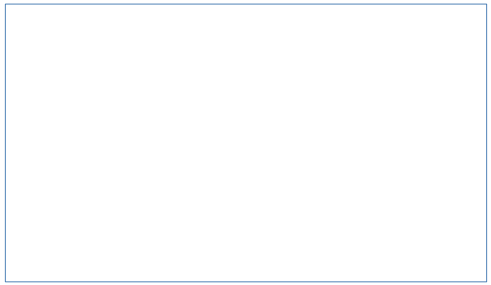
Lecture 24: Transit Orientated Development (TOD)

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Contents

- Transit oriented development: need, objective & benefits
- Principles of TOD
- TOD score & evaluation
- Walk shed & influence zone
- Policy for TOD in India



Transit Oriented Development (TOD)

- TOD is a land-use planning concept.
- Urban development is done along the transit corridor (i.e., MRTS, BRTS etc.)

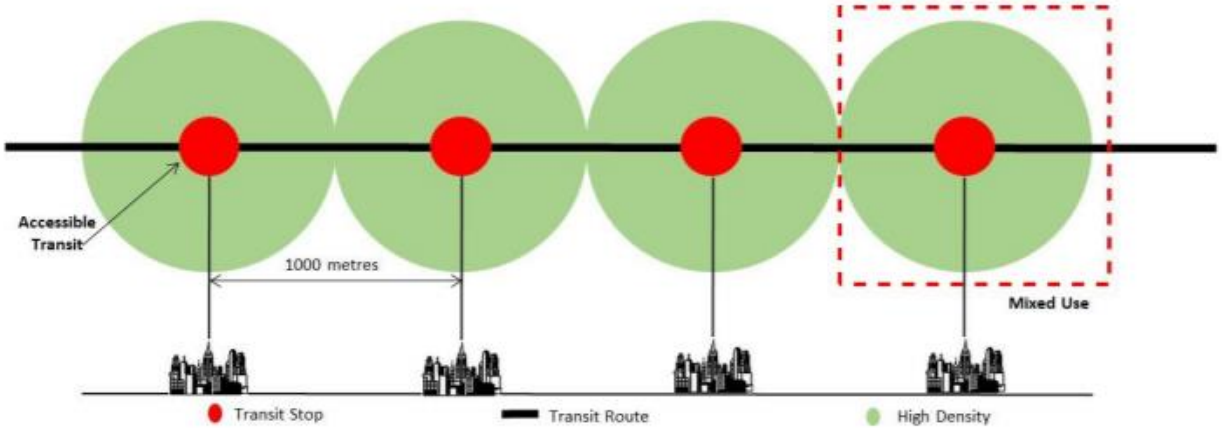


Image: Representing concept of TOD

Source: Ministry of Urban Development

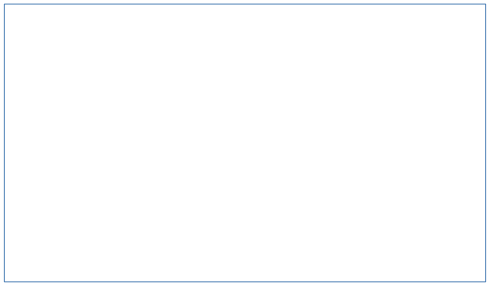
Definition of TOD

- Creation of compact, walkable, pedestrian-oriented, mixed-use communities centered around high-quality train systems
- Regional planning, city revitalization, suburban renewal, and walkable neighborhoods are part & parcel of TOD

TOD promote lower-stress life without complete dependence on a car/personal vehicle for mobility and survival

Factors driving towards TOD

- Rapidly growing traffic congestions nation-wide
- Suburbia
- Strip development
- Demand for more walkable lifestyles away from traffic
- Changes in family structures
- Advocacy for sustainable development



Traffic Congestion: Factors driving towards TOD

- Mind-numbing traffic congestion



- 51000+ vehicle were bought per day in January 2019 in India.
- This is creating more & more congestion on road.

Suburbia: Factors driving towards TOD

Suburbia is lifestyle of sub-urban region which is much lower than urban center.

Issues in suburbia:

- Poor access to public transport
- No major marketplace or entertainment centers
- Less secure neighborhood
- Less job opportunities
- Poor public amenities



Strip development: Factors driving towards TOD

Features of strip development:

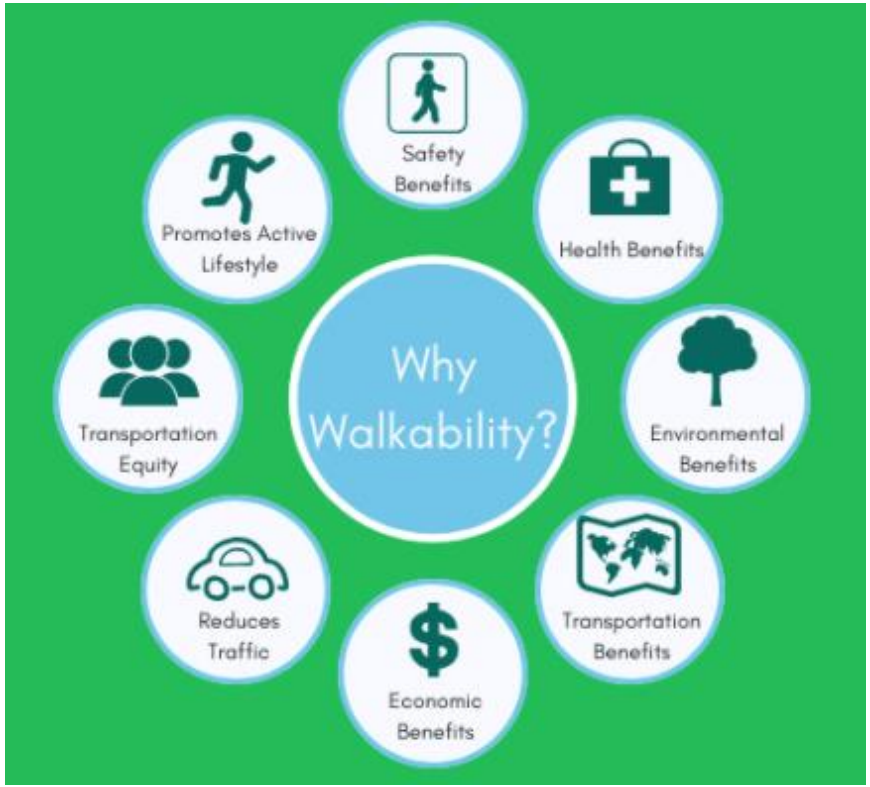
- Broad road frontage
- Predominance of single-story buildings
- Direct access to highway/major road
- Lack of connection to any existing settlement except by highway



- Poor land-use
- Inaccessible for NMT or walk
- Caters few percentage of population

Image source: nextpitsburg.com

Demand for walkable community: Factors driving towards TOD



Source: completecommunitiesde.org

Demand for walkable communities are increasing which ensure higher satisfaction level and multiple benefits to users.

Changes in family structures: Factors driving towards TOD

- With time, nuclear family culture is increasing
- Empty nesters are common now a days (only senior citizen in house)
- In this case, small flats/apartments at urban centers are more suitable in terms of:
  - Security
  - Market/job access
  - Amenities
  - Time saving

Sustainable development : Factors driving towards TOD

- Sustainable development has following major elements:
  - Inclusive
  - Equitable
  - Environment friendly

Following TOD system may fulfill all these three requirement through access and further socio-economic opportunities to various sections of society.

Benefits of TOD- 1/2

- Better access to live, work, and play thus high quality of life
- Greater mobility with ease of moving around
- Increased transit ridership
- Reduced traffic congestion, car accidents and injuries
- Reduced household spending on transportation
- Healthier lifestyle with more walking, and less stress

Benefits of TOD- 2/2

- Higher property values
- Increased foot traffic and customers for area businesses
- Reduced pollution and environmental damage
- Reduced incentive to sprawl, increased incentive for compact development
- Less expensive than building roads and sprawl
- Enhanced ability to maintain economic competitiveness

8 TOD Principles

- Connect
- Walk
- Cycle
- Transit
- Mix
- Compact
- Densify
- Shift



Source: ITDP

TOD Principle-1: Connect

- High level of connectivity at station areas.
- Greater amount of access is granted to the most people.

As each station increases its reach into the larger community, access to the region's economy is improved.



Image: Delhi metro connect all major points in NCR (National Capital Region)

TOD Principle-2: Walk

High quality, unobstructed pedestrian footpaths provide basic mobility for all

Furniture, landscaping elements, and active building edges transform walkways into vibrant public spaces

Safe and complete

Active and vibrant

Temperate and comfortable

TOD Principle-3: Cycle

- Safety for cyclists from motorized traffic
- Creating separate cycle tracks
- Complete network
- Adequate shading elements,
- Separate cycle parking



Image source: makingspaceforcycle.org

TOD Principle-4: Transit

- High capacity to ensure comfortable ride even at peak hours.
- High frequency to minimize the waiting time.

Frequency of Delhi metro at peak hours is 2 to 3 minutes.

8 coach metro has 400 seats and can carry much more passengers.



Image source: business-standard.com

TOD Principle-5: Densify

- To absorb urban growth in compact and dense forms, urban areas must grow vertically (densification) instead of horizontally (sprawl)
- High urban densities oriented towards transit support a transit service of high-quality frequency and connectivity
- Generate resources for investment in system improvements and expansions



TOD Principle-6: Compact



In a compact city, or a compact district

- various activities and uses are conveniently located close together
- minimizing the time and energy required to reach them
- maximizing the potential for interaction

Image source: balancegrowth.ohio.gov

TOD Principle-7: Mix

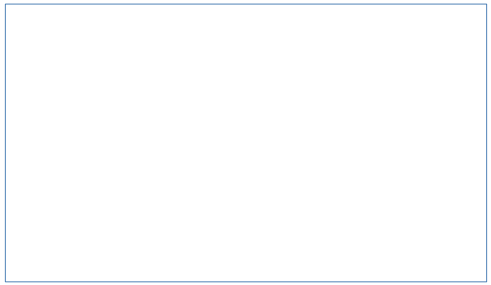
- Provide a balanced mix of complementary uses and activities within proximity
- Increasing the chances that people can reach most of their daily needs by foot, bicycle, or transit
- A strong mix of uses keeps streets active and safe
- Balanced mix of uses and activities provides residents a true choice of lifestyles



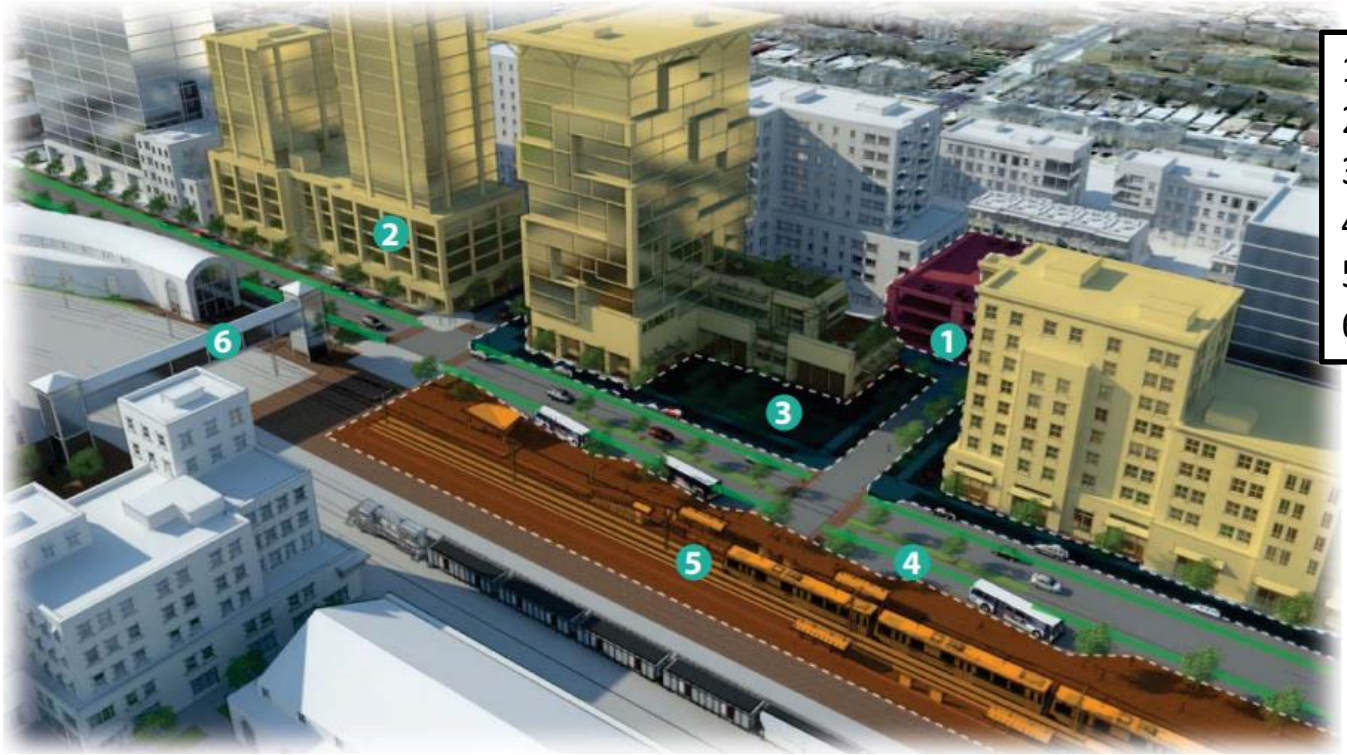
Image source: propertyguru.com.my

TOD Principle-8: Shift

- New way of thinking about personal mobility
- Car dependent city to a multi-modal city
- Goes beyond needed transit improvements (consider socio-economic needs of users)
- Mode shift from car to NMT/PT/pedestrian

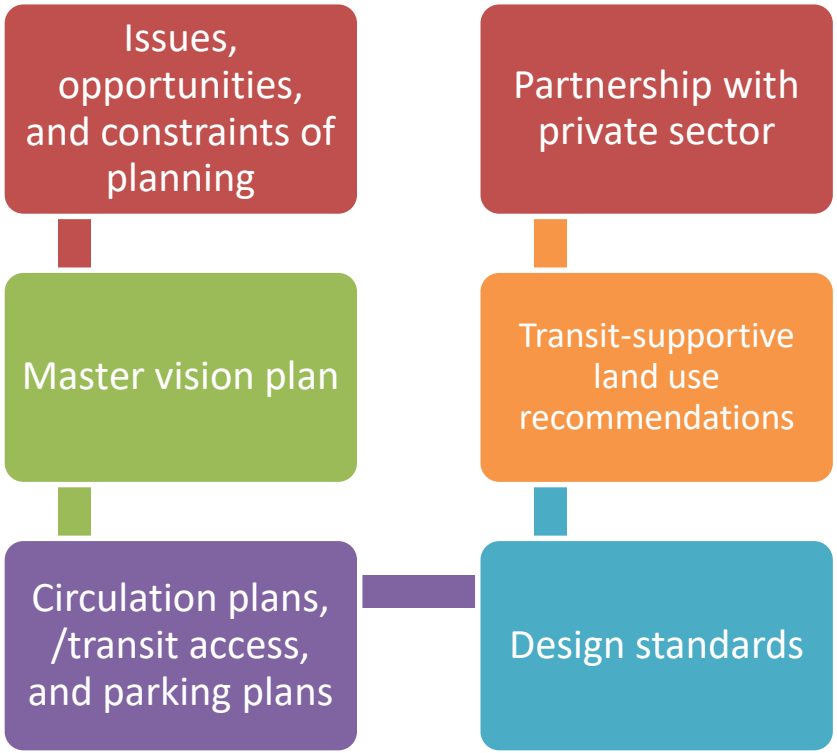


Example of TOD at Urban Center



1- Parking facility
2- Offices/employment centers
3- Open spaces
4- NMT friendly infrastructure
5- High frequency transit
6- Pedestrian Infrastructure

TOD planning process



Features for transit node

- Train station as prominent feature of town center
- Public square fronting train station
- Supporting end mile transport (Feeder buses etc.)
- Bikeshare rental system and bikeway network integrated into stations
- Specialized retail at stations serving commuters and locals including cafes, grocery, dry cleaners



Image source: citizenmatter.in

TOD Score (UN-Habitat & ITDP)

TOD Parameters	Weightage
Walk	15
Cycle	5
Connect	15
Transit	Avg. walking distance from transit (meters)
Mix	25
Densify	15
Compact	10
Shift	15

ITDP: Institute for Transportation and Development Policy

UN (United Nations)-Habitat

Back

- Vision of "a better quality of life for all in an urbanizing world"
- UN-Habitat works with partners to build inclusive, safe, resilient and sustainable cities and communities
- Established in 1978
- Outcome of the first United Nations Conference on Human Settlements and Sustainable Urban Development held in Vancouver, Canada, in 1976

TOD evaluation based on 3-Vs (World Bank)



Node value

- Importance of a station in the public transit network



Place value

- Quality and attractiveness of the area around the station.
- Availability of services
- Pedestrian accessibility



Market potential value

- Unrealized market value of station areas.
- Demand for land

Walkshed

- Transit Oriented Development areas are generally identified by their walkshed
- Area covers the distance assumed people will walk to get to a transit station
- For rail transit, it is higher than bus transit



Improvement required for better walkshed

- Remove highways and highway ramps
- Add pedestrian bridges
- Add off-street trails (cycle & walk exclusive path)



National TOD Policy (India) 2017: Need

- Level of urbanization has increased from 17.29% in 1951 to 31.6 % in 2011 and will grow to 600 million by 2030
- Urbanization has led to horizontal growth of the cities thus creating problems of urban sprawl
- Transit systems like metro rail, BRTS, etc. are being implemented to cater to the growing travel demand.
- It has thus become inevitable to have TOD for all such cities which have an existing mass transit systems or are planning to do so

Source: Ministry of Housing & Urban Affairs

TOD Policy (India): Vision

- Enable Transformation: to assist in transformation of cities from private vehicle dependent city to public transport-oriented development

- Accessible Public Transport: to promote the usage of public transport by making it accessible, encourage green mobility by encouraging people to walk and cycle and at the same time curb pollution and other negative impacts of motorization.

- Compact Walkable Communities: to create livable and affordable communities, which are compact and walkable

TOD component in policy



Supporting tools for TOD implementation



Influence Zone/Walkshed

- Up to a radius of nearly 500-800m of the transit station
- If distance between the transit stations is less than 1 km and overlap in the influence area, it can be identified as a delineated zone (around 500m) on either side of the transit corridor within 10 - 12 minutes walking distance
- Demarcated and notified through master plan and local area plans before implementation of transit



High density

- Densification in the influence area by providing higher Floor Area Ratio (FAR)/ Floor Space Index (FSI).
- Higher population & job density as compared to the area around and beyond the influence areas.
- Minimum FAR should be 300 - 500%, and can be higher, depending on the city size.
- Not necessary to keep the density and FAR norms consistent for the influence areas across the city.

Mandatory inclusive housing

- The cities should fix a minimum percentage (30% or higher) of allowed FAR for affordable housing (for example up to 60 sq.mt. area) in all development/redevelopment in the influence zones.
- Housing in the influence zone should have a mix of all economic groups/ sections.
- The development control regulation should stipulate housing for Economically Weaker Sections (EWS)

Multimodal integration

- The influence area should have high quality integrated multimodal transport system for the optimum use of the facilities by the residents/users.

- Ensure first and last mile connectivity does not become a bottleneck in the use of public transit systems by the citizens

- Stations, should be designed to provide high quality services that assure user satisfaction in terms of safety and comfort.

- The hierarchy of the facilities at the transit system should prioritize pedestrians followed by bicycle, feeder buses, drop-off facilities and park and ride facility in the given order

Conclusion

- Urban centers need to be compact and of mix use with high density.
- TOD is a strong tool through which transit-oriented development can be done which is accessible for residents.
- TOD promotes NMT & walk which support environment friendly transit system.
- Comprehensive guidelines for India is outlines in policy document of ministry of housing & urban affairs.

References

- The Centre for Neighborhood Technology, 2013; “Transit-Oriented Development in the Chicago Region Efficient and Resilient Communities for the 21st Century”

- Denver City Council, 2014 “Transit Oriented Denver-Transit oriented development strategic plan”

- TCRP report, 2008 “Effects of TOD on housing, parking & travel”

Thank You